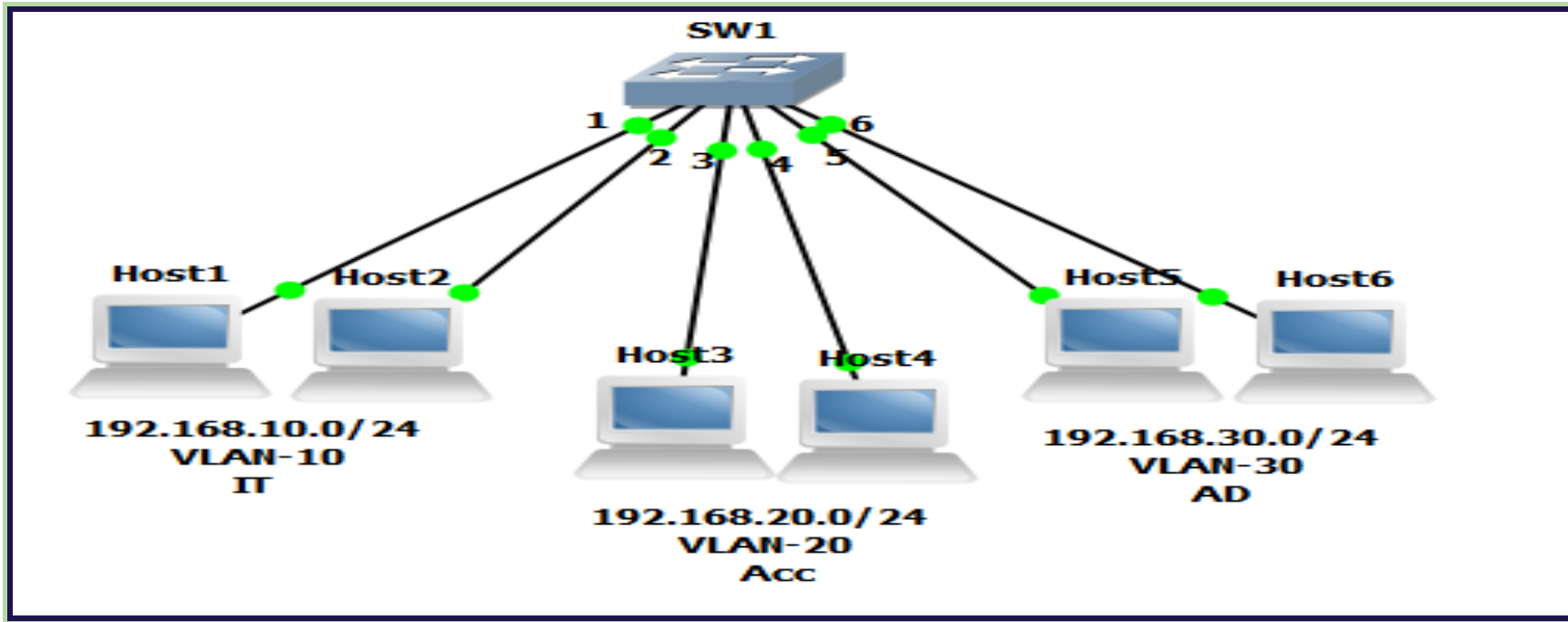


VLAN Configuration

Switch1 VLAN Configuration



```
Switch>enable
```

```
Switch#configure terminal
```

```
Switch(config)#hostname Sw1
```

```
Sw1(config)#no ip domain-lookup
```

```
Sw1(config)#line console 0
```

```
Sw1(config-line)#exec-timeout 0
```

```
Sw1(config-line)#end
```

```
Sw1#write
```

```
Sw1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up
FastEthernet0/3	unassigned	YES	manual	up	up
FastEthernet0/4	unassigned	YES	manual	up	up
FastEthernet0/5	unassigned	YES	manual	up	up
FastEthernet0/6	unassigned	YES	manual	up	up
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down
FastEthernet0/22	unassigned	YES	manual	down	down
FastEthernet0/23	unassigned	YES	manual	down	down
FastEthernet0/24	unassigned	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	manual	down	down
GigabitEthernet0/2	unassigned	YES	manual	down	down
Vlan1	unassigned	YES	manual	administratively down	down

```
Sw1#show running-config
```

```
Building configuration...
```

```
Current configuration : 1077 bytes
```

```
!
```

```
version 12.2
```

```
no service timestamps log datetime msec
```

```
no service timestamps debug datetime msec
```

```
no service password-encryption
```

```
!
```

```
hostname Sw1
```

```
!
```

```
spanning-tree mode pvst
```

```
spanning-tree extend system-id
```

```
!
```

```
interface FastEthernet0/1
```

```
!
```

```
interface FastEthernet0/2
```

```
!
```

```
interface FastEthernet0/3
```

```
!
```

```
interface FastEthernet0/4
```

```
!
```

```
interface FastEthernet0/5
```

```
!
```

```
interface FastEthernet0/6
```

```
!
```

```
interface FastEthernet0/7
```

```
!
```

```
interface FastEthernet0/8
```

```
!
```

```
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
```

```

!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
  no ip address
  shutdown
!
line con 0
!
line vty 0 4
  login
line vty 5 15
  login
!
End

```

```
Sw1#show vlan brief
```

VLAN Name	Status	Ports
-----	-----	-----
1 default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	

```
1005 trnet-default      active
```

```
Sw1#configure terminal
```

```
Sw1(config)#vlan ?
```

```
<1-4094> ISL VLAN IDs 1-1005
```

```
Sw1(config)#vlan 10
```

```
Sw1(config-vlan)#name IT
```

```
Sw1(config-vlan)#exit
```

```
Sw1(config)#vlan 20
```

```
Sw1(config-vlan)#name Acc
```

```
Sw1(config-vlan)#exit
```

```
Sw1(config)#vlan 30
```

```
Sw1(config-vlan)#name AD
```

```
Sw1(config-vlan)#end
```

```
Sw1#write
```

```
Sw1#show vlan brief
```

VLAN Name		Status	Ports
-----		-----	-----
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	IT	active	
20	Acc	active	

30	AD	active
1002	fddi-default	active
1003	token-ring-default	active
1004	fddinet-default	active
1005	trnet-default	active

```
Sw1#configure terminal
```

```
Sw1(config)#interface fastEthernet 0/1
```

```
Sw1(config-if)#switchport mode access
```

```
Sw1(config-if)#switchport access vlan 10
```

```
Sw1(config-if)#exit
```

```
Sw1(config)#interface fastEthernet 0/2
```

```
Sw1(config-if)#switchport mode access
```

```
Sw1(config-if)#switchport access vlan 10
```

```
Sw1(config-if)#exit
```

```
Sw1(config)#interface range fastEthernet 0/3-4
```

```
Sw1(config-if-range)#switchport mode access
```

```
Sw1(config-if-range)#switchport access vlan 20
```

```
Sw1(config-if-range)#exit
```

```
Sw1(config)#interface range fastEthernet 0/5-6
```

```
Sw1(config-if-range)#switchport mode access
```

```
Sw1(config-if-range)#switchport access vlan 30
```

```
Sw1(config-if-range)#end
```

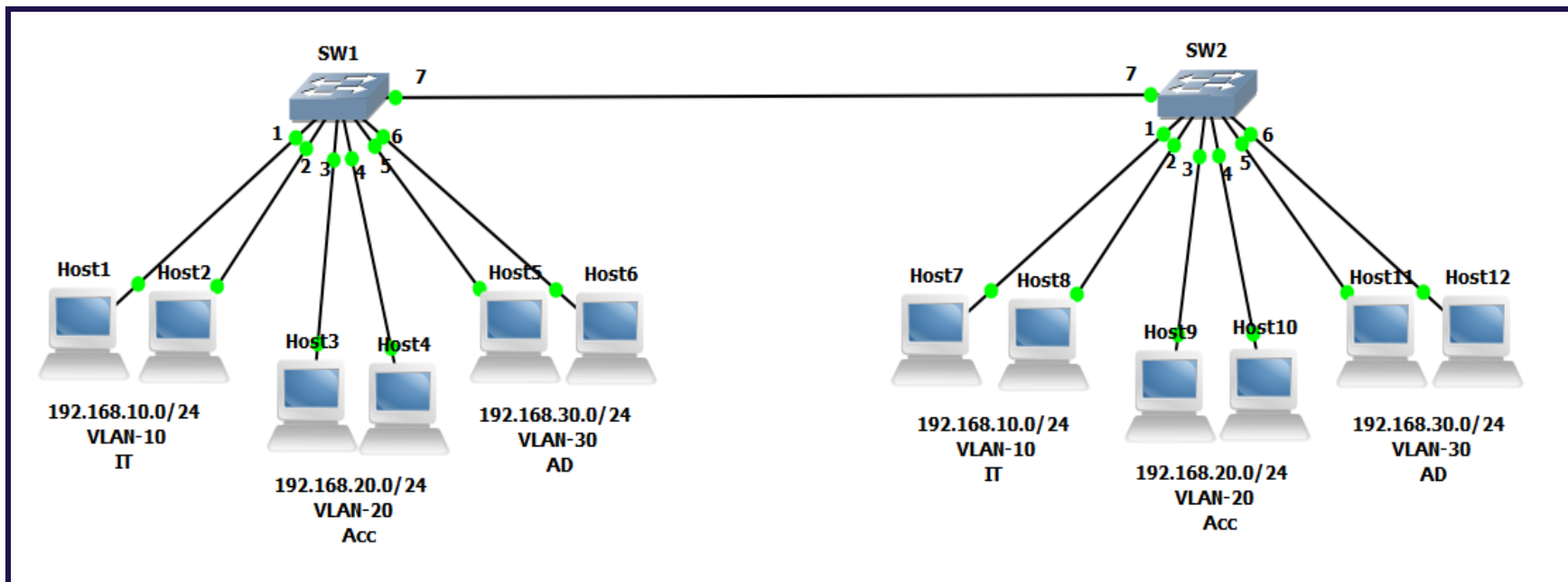
```
Sw1#write
```

```
Sw1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10

			Fa0/11, Fa0/12, Fa0/13, Fa0/14
			Fa0/15, Fa0/16, Fa0/17, Fa0/18
			Fa0/19, Fa0/20, Fa0/21, Fa0/22
			Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	IT	active	Fa0/1, Fa0/2
20	Acc	active	Fa0/3, Fa0/4
30	AD	active	Fa0/5, Fa0/6
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Switch2 VLAN Configuration



```
Switch>enable
```

```
Switch#configure terminal
```

```
Switch(config)#hostname Sw2
```

```
Sw2(config)#no ip domain-lookup
```

```
Sw2(config)#line console 0
```

```
Sw2(config-line)#exec-timeout 0
```

```
Sw2(config-line)#end
```

```
Sw2#write
```

```
Sw2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Sw2#configure terminal
```

```
Sw2(config)#vlan 10
```

```
Sw2(config-vlan)#name IT
```

```
Sw2(config-vlan)#exit
```

```
Sw2(config)#vlan 20
```

```
Sw2(config-vlan)#name Acc
```

```
Sw2(config-vlan)#exit
```

```
Sw2(config)#vlan 30
Sw2(config-vlan)#name AD
Sw2(config-vlan)#end
Sw2#write
```

```
Sw2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	IT	active	
20	Acc	active	
30	AD	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Sw2#configure terminal
Sw2(config)#interface range fastEthernet 0/1-2
Sw2(config-if-range)#switchport mode access
Sw2(config-if-range)#switchport access vlan 10
Sw2(config-if-range)#exit
```

```
Sw2(config)#interface range fastEthernet 0/3-4
Sw2(config-if-range)#switchport mode access
```

```
Sw2(config-if-range)#switchport access vlan 20
```

```
Sw2(config-if-range)#exit
```

```
Sw2(config)#interface range fastEthernet 0/5-6
```

```
Sw2(config-if-range)#switchport mode access
```

```
Sw2(config-if-range)#switchport access vlan 30
```

```
Sw2(config-if-range)#end
```

```
Sw2#write
```

```
Sw2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	IT	active	Fa0/1, Fa0/2
20	Acc	active	Fa0/3, Fa0/4
30	AD	active	Fa0/5, Fa0/6
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Trunk Configuration

```
Host1 C:\>ping 192.168.10.2
```

```
Pinging 192.168.10.2 with 32 bytes of data:
```

```
Reply from 192.168.10.2: bytes=32 time=1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=2ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.10.2:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

```
Host1 C:\>ping 192.168.10.3
```

```
Pinging 192.168.10.3 with 32 bytes of data:
```

```
Request timed out.
```

```
Request timed out.
```

```
Request timed out.
```

```
Request timed out.
```

```
Ping statistics for 192.168.10.3:
```

```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
Sw1#show interfaces fastEthernet 0/7 switchport
```

```
Name: Fa0/7
```

```
Switchport: Enabled
```

```
Administrative Mode: dynamic auto
```

```
Operational Mode: static access
```

```
Administrative Trunking Encapsulation:
```

```
Operational Trunking Encapsulation: native
```

```
Negotiation of Trunking: On
```

Access Mode VLAN: 1 (default)

Trunking Native Mode VLAN: 1 (default)

Voice VLAN: none

Administrative private-vlan host-association: none

Administrative private-vlan mapping: none

Administrative private-vlan trunk native VLAN: none

Administrative private-vlan trunk encapsulation: dot1q

Administrative private-vlan trunk normal VLANs: none

Administrative private-vlan trunk private VLANs: none

Operational private-vlan: none

Trunking VLANs Enabled: All

Pruning VLANs Enabled: 2-1001

Capture Mode Disabled

Capture VLANs Allowed: ALL

Protected: false

Unknown unicast blocked: disabled

Unknown multicast blocked: disabled

Appliance trust: none

Sw2#show interfaces fastEthernet 0/7 switchport

Name: Fa0/7

Switchport: Enabled

Administrative Mode: dynamic auto

Operational Mode: static access

Administrative Trunking Encapsulation:

Operational Trunking Encapsulation: native

Negotiation of Trunking: On

Access Mode VLAN: 1 (default)

Trunking Native Mode VLAN: 1 (default)

Voice VLAN: none

Administrative private-vlan host-association: none

```
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
```

```
Sw1#configure terminal
Sw1(config)#interface fastEthernet 0/7
Sw1(config-if)#switchport mode trunk
Sw1(config-if)#switchport trunk allowed vlan all
Sw1(config-if)#end
Sw1#write
```

```
Sw2#configure terminal
Sw2(config)#interface fastEthernet 0/7
Sw2(config-if)#switchport mode trunk
Sw2(config-if)#switchport trunk allowed vlan all
Sw2(config-if)#end
Sw2#write
```

```
Sw1#show interfaces fastEthernet 0/7 switchport
Name: Fa0/7
```

Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation:
Operational Trunking Encapsulation: native
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none

```
Sw2#show interfaces fastEthernet 0/7 switchport
```

Name: Fa0/7
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation:

Operational Trunking Encapsulation: native
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none

Host1 C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

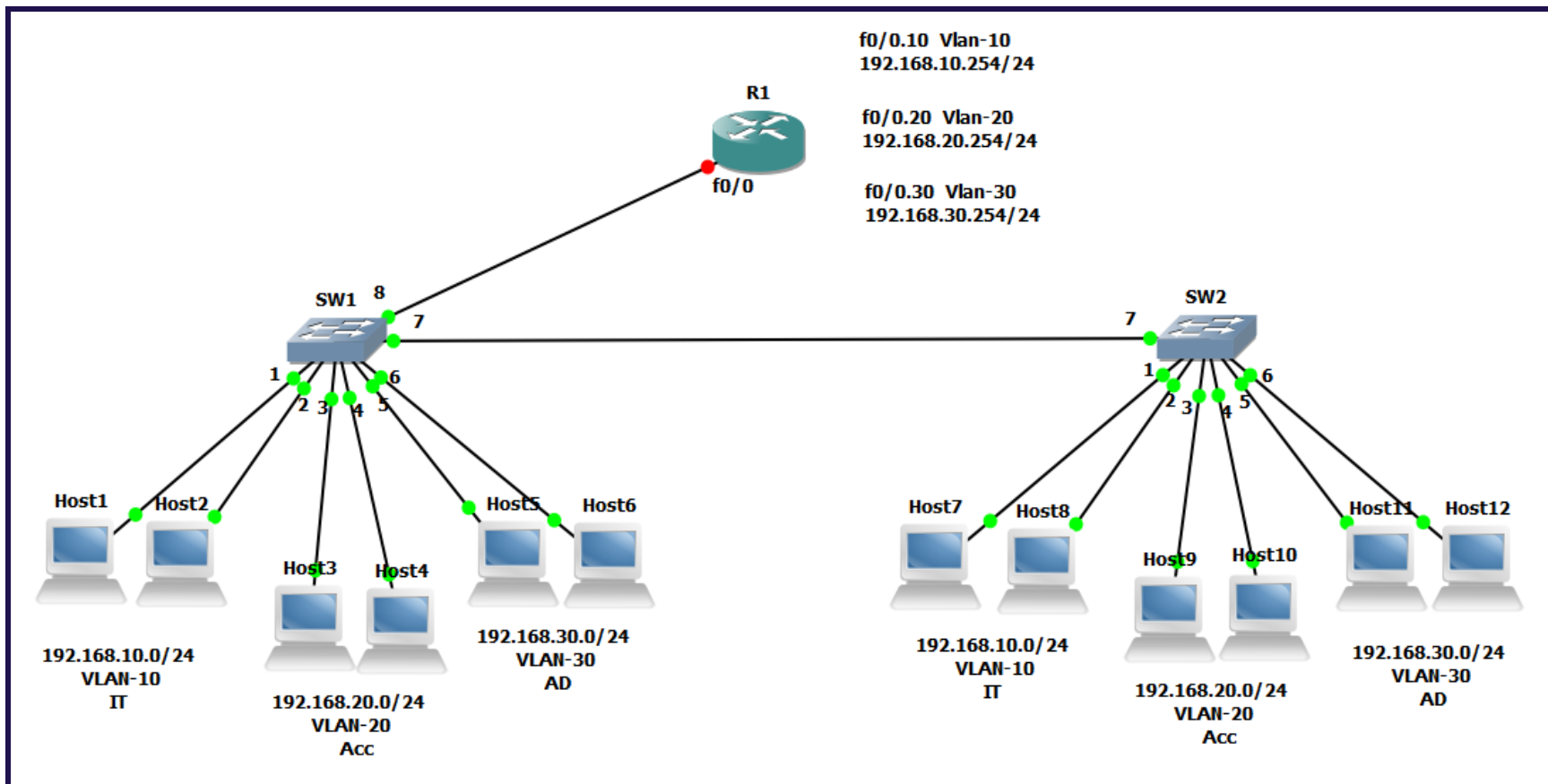
Ping statistics for 192.168.10.3:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

Inter VLAN Configuration



```
Sw1#configure terminal
```

```
Sw1(config)#interface fastEthernet 0/8
```

```
Sw1(config-if)#switchport mode trunk
```

```
Sw1(config-if)#switchport trunk allowed vlan all
```

```
Sw1(config-if)#end
```

```
Sw1#write
```

```
Router>enable
Router#configure terminal
Router(config)#hostname R1
```

```
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#no shutdown
R1(config-if)#exit
```

```
R1(config)#interface gigabitEthernet 0/0.?
<0-4294967295> GigabitEthernet interface number
```

```
R1(config)#interface gigabitEthernet 0/0.10
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.254 255.255.255.0
R1(config-subif)#exit
```

```
R1(config)#interface gigabitEthernet 0/0.20
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.20.254 255.255.255.0
R1(config-subif)#exit
```

```
R1(config)#interface gigabitEthernet 0/0.30
R1(config-subif)#encapsulation dot1Q 30
R1(config-subif)#ip address 192.168.30.254 255.255.255.0
R1(config-subif)#end
R1#write
```

```
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up

GigabitEthernet0/0.10	192.168.10.254	YES	manual	up	up
GigabitEthernet0/0.20	192.168.20.254	YES	manual	up	up
GigabitEthernet0/0.30	192.168.30.254	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

R1#show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
C 192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
C 192.168.30.0/24 is directly connected, GigabitEthernet0/0.30

Sw1#show interfaces fastEthernet 0/8 switchport

Name: Fa0/8

Switchport: Enabled

Administrative Mode: trunk

Operational Mode: trunk

Administrative Trunking Encapsulation:

Operational Trunking Encapsulation: native

Negotiation of Trunking: On

Access Mode VLAN: 1 (default)

Trunking Native Mode VLAN: 1 (default)

Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none

Host1 C:\>ping 192.168.20.4

Pinging 192.168.20.4 with 32 bytes of data:

Request timed out.

Reply from 192.168.20.4: bytes=32 time=1ms TTL=127

Reply from 192.168.20.4: bytes=32 time=1ms TTL=127

Reply from 192.168.20.4: bytes=32 time=2ms TTL=127

Ping statistics for 192.168.20.4:

Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 2ms, Average = 1ms

```
Host1 C:\>ping 192.168.30.4
```

```
Pinging 192.168.30.4 with 32 bytes of data:
```

```
Request timed out.
```

```
Reply from 192.168.30.4: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.30.4: bytes=32 time=1ms TTL=127
```

```
Reply from 192.168.30.4: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 192.168.30.4:
```

```
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
##### THE END #####
```